

use 3 significant figure answers for drawing conclusions.

2(a)	Let the mass of the satellite be m. Gravitational force on satellite by the planet provides centripetal force for the satellite to be in a circular orbit around the planet, with the radius of the orbit given by $(R + h)$, i.e. $\frac{GMm}{(R + h)^2} = \frac{mv_s^2}{(R + h)} \rightarrow v_s^2 = \frac{GM}{R + h}$ So the speed of the satellite, $v_s = \sqrt{\frac{GM}{R + h}}$
2(b)	Let the mass of the rock be m. Initially, at an infinite distance away, the kinetic energy (KE) and the gravitational potential energy (PE) of the rock is zero. Gain in KE of rock = loss in PE of rock, i.e. $\frac{1}{2}mV^2 - 0 = 0 - \left(-\frac{GMm}{R + h}\right) \rightarrow V^2 = \frac{2GM}{R + h}$ So the speed of the rock, $V = \sqrt{\frac{2GM}{R + h}}$
2(c)	If $V < v_s$, the rock would fall to the planet's surface. If $V = v_s$, the rock would go into orbit around the planet. Since $V > v_s$, the rock would travel off into space.
3(a)(i)	Using the usual notation.